$10n^3 = 10^7 \times 1000$ yields $n = 1000$, indicating that the polynomial time algorithm allows us to solve much larger problems.

The point of view emerging from the above discussion is that, as a first cut, it is useful to juxtapose polynomial and exponential time algorithms, the former being viewed as relatively fast and efficient, and the latter as relatively slow. This point of view is justified in many – but not all – contexts and we will be returning to it later in this book.

## 1.7 Exercises

**Exercise 1.1*** Suppose that a function $f : \Re^n \mapsto \Re$ is both concave and convex. Prove that $f$ is an affine function.

**Exercise 1.2** Suppose that $f_1, \ldots, f_m$ are convex functions from $\Re^n$ into $\Re$ and let $f(\mathbf{x}) = \sum_{i=1}^m f_i(\mathbf{x})$.

**(a)** Show that if each $f_i$ is convex, so is $f$.

**(b)** Show that if each $f_i$ is piecewise linear and convex, so is $f$.

**Exercise 1.3** Consider the problem of minimizing a cost function of the form $\mathbf{c}'\mathbf{x} + f(\mathbf{d}'\mathbf{x})$, subject to the linear constraints $\mathbf{A}\mathbf{x} \geq \mathbf{b}$. Here, $\mathbf{d}$ is a given vector and the function $f : \Re \mapsto \Re$ is as specified in Figure 1.8. Provide a linear programming formulation of this problem.

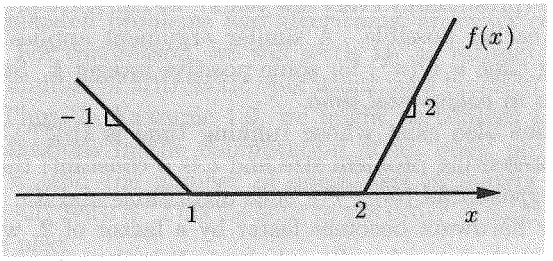


**Figure 1.8:** The function $f$ of Exercise 1.3.

**Exercise 1.4** Consider the problem

$$\begin{array}{ll} \text{minimize} & 2x_1 + 3|x_2 - 10| \\ \text{subject to} & |x_1 + 2| + |x_2| \leq 5, \end{array}$$

and reformulate it as a linear programming problem.

**Exercise 1.5** Consider a linear optimization problem, with absolute values, of the following form:

$$\begin{aligned} \text{minimize} \quad & \mathbf{c}'\mathbf{x} + \mathbf{d}'\mathbf{y} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} \leq \mathbf{b} \\ & y_i = |x_i|, \qquad \forall\, i. \end{aligned}$$

Assume that all entries of $\mathbf{B}$ and $\mathbf{d}$ are nonnegative.

(a) Provide two different linear programming formulations, along the lines discussed in Section 1.3.

(b) Show that the original problem and the two reformulations are equivalent in the sense that either all three are infeasible, or all three have the same optimal cost.

(c) Provide an example to show that if $\mathbf{B}$ has negative entries, the problem may have a local minimum that is not a global minimum. (It will be seen in Chapter 2 that this is never the case in linear programming problems. Hence, in the presence of such negative entries, a linear programming reformulation is implausible.)

**Exercise 1.6** Provide linear programming formulations of the two variants of the rocket control problem discussed at the end of Section 1.3.

**Exercise 1.7 (The moment problem)** Suppose that $Z$ is a random variable taking values in the set $0, 1, \ldots, K$, with probabilities $p_0, p_1, \ldots, p_K$, respectively. We are given the values of the first two moments $E[Z] = \sum_{k=0}^{K} kp_k$ and $E[Z^2] = \sum_{k=0}^{K} k^2 p_k$ of $Z$, and we would like to obtain upper and lower bounds on the value of the fourth moment $E[Z^4] = \sum_{k=0}^{K} k^4 p_k$ of $Z$. Show how linear programming can be used to approach this problem.

**Exercise 1.8 (Road lighting)** Consider a road divided into $n$ segments that is illuminated by $m$ lamps. Let $p_j$ be the power of the $j$th lamp. The illumination $I_i$ of the $i$th segment is assumed to be $\sum_{j=1}^{m} a_{ij} p_j$, where $a_{ij}$ are known coefficients. Let $I_i^*$ be the desired illumination of road $i$.

We are interested in choosing the lamp powers $p_j$ so that the illuminations $I_i$ are close to the desired illuminations $I_i^*$. Provide a reasonable linear programming formulation of this problem. Note that the wording of the problem is loose and there is more than one possible formulation.

**Exercise 1.9** Consider a school district with $I$ neighborhoods, $J$ schools, and $G$ grades at each school. Each school $j$ has a capacity of $C_{jg}$ for grade $g$. In each neighborhood $i$, the student population of grade $g$ is $S_{ig}$. Finally, the distance of school $j$ from neighborhood $i$ is $d_{ij}$. Formulate a linear programming problem whose objective is to assign all students to schools, while minimizing the total distance traveled by all students. (You may ignore the fact that numbers of students must be integer.)

**Exercise 1.10 (Production and inventory planning)** A company must deliver $d_i$ units of its product at the end of the $i$th month. Material produced during

a month can be delivered either at the end of the same month or can be stored as inventory and delivered at the end of a subsequent month; however, there is a storage cost of $c_1$ dollars per month for each unit of product held in inventory. The year begins with zero inventory. If the company produces $x_i$ units in month $i$ and $x_{i+1}$ units in month $i+1$, it incurs a cost of $c_2|x_{i+1} - x_i|$ dollars, reflecting the cost of switching to a new production level. Formulate a linear programming problem whose objective is to minimize the total cost of the production and inventory schedule over a period of twelve months. Assume that inventory left at the end of the year has no value and does not incur any storage costs.

**Exercise 1.11 (Optimal currency conversion)** Suppose that there are $N$ available currencies, and assume that one unit of currency $i$ can be exchanged for $r_{ij}$ units of currency $j$. (Naturally, we assume that $r_{ij} > 0$.) There also certain regulations that impose a limit $u_i$ on the total amount of currency $i$ that can be exchanged on any given day. Suppose that we start with $B$ units of currency 1 and that we would like to maximize the number of units of currency $N$ that we end up with at the end of the day, through a sequence of currency transactions. Provide a linear programming formulation of this problem. Assume that for any sequence $i_1, \ldots, i_k$ of currencies, we have $r_{i_1 i_2} r_{i_2 i_3} \cdots r_{i_{k-1} i_k} r_{i_k i_1} \leq 1$, which means that wealth cannot be multiplied by going through a cycle of currencies.

**Exercise 1.12 (Chebychev center)** Consider a set $P$ described by linear inequality constraints, that is, $P = \{\mathbf{x} \in \Re^n \mid \mathbf{a}_i'\mathbf{x} \leq b_i,\ i = 1, \ldots, m\}$. A ball with center $\mathbf{y}$ and radius $r$ is defined as the set of all points within (Euclidean) distance $r$ from $\mathbf{y}$. We are interested in finding a ball with the largest possible radius, which is entirely contained within the set $P$. (The center of such a ball is called the *Chebychev center* of $P$.) Provide a linear programming formulation of this problem.

**Exercise 1.13 (Linear fractional programming)** Consider the problem

$$\begin{array}{ll} \text{minimize} & \dfrac{\mathbf{c}'\mathbf{x} + d}{\mathbf{f}'\mathbf{x} + g} \\ \text{subject to} & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{f}'\mathbf{x} + g > 0. \end{array}$$

Suppose that we have some prior knowledge that the optimal cost belongs to an interval $[K, L]$. Provide a procedure that uses linear programming as a subroutine and allows us to compute the optimal cost within any desired accuracy. *Hint:* Consider the problem of deciding whether the optimal cost is less than or equal to a certain number.

**Exercise 1.14** A company produces and sells two different products. The demand for each product is unlimited, but the company is constrained by cash availability and machine capacity.

Each unit of the first and second product requires 3 and 4 machine hours, respectively. There are 20,000 machine hours available in the current production period. The production costs are \$3 and \$2 per unit of the first and second product, respectively. The selling prices of the first and second product are \$6 and \$5.40 per unit, respectively. The available cash is \$4,000; furthermore, 45%

of the sales revenues from the first product and 30% of the sales revenues from the second product will be made available to finance operations during the current period.

(a) Formulate a linear programming problem that aims at maximizing net income subject to the cash availability and machine capacity limitations.

(b) Solve the problem graphically to obtain an optimal solution.

(c) Suppose that the company could increase its available machine hours by 2,000, after spending $400 for certain repairs. Should the investment be made?

**Exercise 1.15** A company produces two kinds of products. A product of the first type requires 1/4 hours of assembly labor, 1/8 hours of testing, and $1.2 worth of raw materials. A product of the second type requires 1/3 hours of assembly, 1/3 hours of testing, and $0.9 worth of raw materials. Given the current personnel of the company, there can be at most 90 hours of assembly labor and 80 hours of testing, each day. Products of the first and second type have a market value of $9 and $8, respectively.

(a) Formulate a linear programming problem that can be used to maximize the daily profit of the company.

(b) Consider the following two modifications to the original problem:

(i) Suppose that up to 50 hours of overtime assembly labor can be scheduled, at a cost of $7 per hour.

(ii) Suppose that the raw material supplier provides a 10% discount if the daily bill is above $300.

Which of the above two elements can be easily incorporated into the linear programming formulation and how? If one or both are not easy to incorporate, indicate how you might nevertheless solve the problem.

**Exercise 1.16** A manager of an oil refinery has 8 million barrels of crude oil A and 5 million barrels of crude oil B allocated for production during the coming month. These resources can be used to make either gasoline, which sells for $38 per barrel, or home heating oil, which sells for $33 per barrel. There are three production processes with the following characteristics:

| | Process 1 | Process 2 | Process 3 |
|---|---|---|---|
| Input crude A | 3 | 1 | 5 |
| Input crude B | 5 | 1 | 3 |
| Output gasoline | 4 | 1 | 3 |
| Output heating oil | 3 | 1 | 4 |
| Cost | $51 | $11 | $40 |

All quantities are in barrels. For example, with the first process, 3 barrels of crude A and 5 barrels of crude B are used to produce 4 barrels of gasoline and

3 barrels of heating oil. The costs in this table refer to variable and allocated overhead costs, and there are no separate cost items for the cost of the crudes. Formulate a linear programming problem that would help the manager maximize net revenue over the next month.

**Exercise 1.17 (Investment under taxation)** An investor has a portfolio of $n$ different stocks. He has bought $s_i$ shares of stock $i$ at price $p_i$, $i = 1, \ldots, n$. The current price of one share of stock $i$ is $q_i$. The investor expects that the price of one share of stock $i$ in one year will be $r_i$. If he sells shares, the investor pays transaction costs at the rate of 1% of the amount transacted. In addition, the investor pays taxes at the rate of 30% on capital gains. For example, suppose that the investor sells 1,000 shares of a stock at \$50 per share. He has bought these shares at \$30 per share. He receives \$50,000. However, he owes $0.30 \times (50{,}000 - 30{,}000) = \$6{,}000$ on capital gain taxes and $0.01 \times (50{,}000) = \$500$ on transaction costs. So, by selling 1,000 shares of this stock he nets $50{,}000 - 6{,}000 - 500 = \$43{,}500$. Formulate the problem of selecting how many shares the investor needs to sell in order to raise an amount of money $K$, net of capital gains and transaction costs, while maximizing the expected value of his portfolio next year.

**Exercise 1.18** Show that the vectors in a given finite collection are linearly independent if and only if none of the vectors can be expressed as a linear combination of the others.

**Exercise 1.19** Suppose that we are given a set of vectors in $\Re^n$ that form a basis, and let $\mathbf{y}$ be an arbitrary vector in $\Re^n$. We wish to express $\mathbf{y}$ as a linear combination of the basis vectors. How can this be accomplished?

**Exercise 1.20**

(a) Let $S = \{\mathbf{Ax} \mid \mathbf{x} \in \Re^n\}$, where $\mathbf{A}$ is a given $m \times n$ matrix. Show that $S$ is a subspace of $\Re^m$.

(b) Assume that $S$ is a proper subspace of $\Re^n$. Show that there exists a matrix $\mathbf{B}$ such that $S = \{\mathbf{y} \in \Re^n \mid \mathbf{By} = \mathbf{0}\}$. *Hint:* Use vectors that are orthogonal to $S$ to form the matrix $\mathbf{B}$.

(c) Suppose that $V$ is an $m$-dimensional affine subspace of $\Re^n$, with $m < n$. Show that there exist linearly independent vectors $\mathbf{a}_1, \ldots, \mathbf{a}_{n-m}$, and scalars $b_1, \ldots, b_{n-m}$, such that

$$V = \left\{\mathbf{y} \mid \mathbf{a}_i'\mathbf{y} = b_i,\ i = 1, \ldots, n-m\right\}.$$

## 1.8 History, notes, and sources

The word "programming" has been used traditionally by planners to describe the process of operations planning and resource allocation. In the 1940s, it was realized that this process could often be aided by solving optimization problems involving linear constraints and linear objectives. The term "linear programming" then emerged. The initial impetus came in the aftermath of World War II, within the context of military planning problems. In 1947, Dantzig proposed an algorithm, the *simplex method*, which